

The Kojin

Specification and Deployment Guide

SubThought Corporation

Version 1.0 — August 2026

Table of Contents

1. Overview
 2. The Percept Relation
 3. Percepts by Channel
 4. Common Content Idiom Slots
 5. Actuations
 6. Cross-Cutting Structure
 7. The Jungian Model
 8. Architecture
 9. Deployment File Set
 10. The Package
 11. The Theory
 12. Configuration
 13. Test Suites
 14. Privacy and Consent
 15. Domain Coverage
 16. Etymology
-

1 Overview

The Kojin is a Psyche: the interface between a GIL mind and a single human being as world. It mediates two flows — percepts traveling inward from the person's expressed life to the mind's mechanisms, and actuations traveling outward from the mind's mechanisms back to the person. Where the Aether interfaces with the internet-as-world, the Expanse with

physical-reality-as-world, and the Eidos with a grid-as-world, the Kojin interfaces with the human-as-world.

The human-as-world is not a bounded simulation and not a passive data source. It is a living subject who converses, journals, shares documents, expresses preferences, reveals strengths and weaknesses, and moves through a landscape of opportunities and threats. The Kojin collects this personal data — always with consent — and helps the mind build, refine, and act on a model of the person. That model has two registers: the observable life record (biography, preferences, capabilities, circumstances) and the depth-psychological structure (a working model of the person's Jungian ego and Jungian shadow, inferred from what the person says, avoids, repeats, projects, and dreams).

The Kojin serves four functions: **learn** the person's preferences, strengths, and weaknesses; **guide** the person toward opportunities and away from threats; **journal** with the person through conversation and analysis of shared information; and **model** the person's ego and shadow so guidance is grounded in who the person actually is, not just what they say they want.

Unlike the Eidos (turn-based) but like the Aether, the Kojin is event-driven and slow-polling: percepts arrive when the person speaks, writes, or shares, and on periodic review of accumulated material.

All percepts from the Kojin carry `:Modality Kojin`. The `:Channel` slot selects the domain of personhood. The `:Data` slot names the parsable structure type. The `:Content` slot carries the idiom.

2 The Percept Relation

The Percept relation from `totality.theory` is canonical. It does not change for the Kojin.

```

(relation Percept
  :M                ; lexified monad
  :Modality         ; which Psyche – Kojin for this device
  :Channel          ; sub-channel within the Psyche
  :Address nil      ; source URI of the data
  :Data nil         ; data format or parsable structure type
  :Content nil      ; idiom { :Slot value ... } – the actual structured
data
)

(template Percept
{
  :Modality nil
  :Channel nil
  :Address nil
  :Data nil
  :Content nil
  :Moment 0      ; when the percept was received
})

```

2.1 Slot Semantics

Slot	Meaning	Kojin Examples
:Modality	Which Psyche this came through. The Psyche is the sense organ.	Kojin (always)
:Channel	Sub-channel within the Psyche. Domain of personhood.	Dialogue, Journal, Document, Preference, Capability, Situation, Depth, System
:Address	Source URI of the data. Where it came from.	"kojin:///alex/session-14", "file:///uploads/resume.pdf", "gmail:///alex/msg-9921", nil

:Data	Data format or parsable structure type. What the :Content contains.	utterance , journal-entry , document-shared , preference-signal , strength-observed , weakness-observed , opportunity-detected , threat-detected , ego-trait , shadow-trait
:Content	Idiom <code>{:Slot value ...}</code> with the actual structured data.	<code>{:Text "... " :Sentiment -0.3 :Topics {work family} :SessionId "s14"}</code>
:Moment	When the percept was received. (Template slot.)	<code>\@m{20260808000000000}</code>

For comparison, the Aether's channels partition the internet (Web, FileSystem, Message); the Expanse's channels partition the sensorium (Visual, Auditory, Haptic). The Kojin's channels partition the *person*: what they say (Dialogue), what they write reflectively (Journal), what they share (Document), what they want (Preference), what they can and cannot do (Capability), what surrounds them (Situation), and who they are beneath the surface (Depth).

3 Percepts by **:Channel**

All percepts from the Kojin have **:Modality Kojin**. The **:Channel** identifies the domain, **:Data** identifies the format, **:Address** identifies the source, and **:Content** carries the structured data as an idiom.

3.1 Channel Dialogue

:Data	:Address	:Content (idiom)
utterance	"kojin://alex/session-14"	<code>{:Text "I turned down the promotion" :SessionId "s14" :ThreadId "t2" :Sentiment -0.3 :Topics {work} :Attachments {}}</code>

question-answered	"kojin://alex/session-14"	{:QuestionId "q-88" :Text "...":Evasive no :Latency 4200}
question-avoided	"kojin://alex/session-14"	{:QuestionId "q-89" :Topic family :AvoidanceType deflection}

3.2 Channel Journal

:Data	:Address	:Content (idiom)
journal-entry	"kojin://alex/journal/2026-08-08"	{:Text "...":WordCount 340 :Sentiment 0.1 :Topics {sleep career} :Prompted yes :PromptId "p-12"}
journal-pattern	nil	{:PatternType recurring-theme :Theme "fear of being ordinary" :Occurrences 7 :FirstSeen \@m{...} :LastSeen \@m{...}}
dream-report	"kojin://alex/journal/2026-08-08"	{:Text "...":Figures {pursuer water house} :Affect anxious}

3.3 Channel Document

:Data	:Address	:Content (idiom)
document-shared	"file:///uploads/resume.pdf"	{:ContentType "application/pdf" :Size 88213 :DocumentType resume :Summary "...":Entities {...}}
message-shared	"gmail://alex/msg-9921"	{:Counterparty "boss@corp.com" :Direction received :Summary "...":Tone terse}
record-shared	"file:///uploads/lab-results.pdf"	{:RecordType medical :Summary "...":Sensitivity high}

3.4 Channel Preference

:Data	:Address	:Content (idiom)
preference-signal	nil	{:Domain food :Object "cilantro" :Valence -0.9 :Source stated :Confidence 0.95}
preference-revealed	nil	{:Domain work :Object "remote work" :Valence 0.7 :Source behavioral :Evidence {...} :Confidence 0.6}
value-signal	nil	{:Value autonomy :Rank 1 :Source inferred :Evidence {...}}

Stated preferences come from the person's words; revealed preferences come from their behavior. The two frequently disagree — that disagreement is itself a Depth percept (§ 3.7).

3.5 Channel Capability

:Data	:Address	:Content (idiom)
strength-observed	nil	{:Skill "public speaking" :Evidence {...} :Confidence 0.8 :Source {dialogue document}}
weakness-observed	nil	{:Skill "delegation" :Evidence {...} :Confidence 0.7 :Impact high}
growth-signal	nil	{:Skill "spanish" :Direction improving :Delta 0.1 :Window \@m{90}}

3.6 Channel Situation

:Data	:Address	:Content (idiom)
opportunity-detected	"https://jobs..."	{:Kind career :Description "..." :FitScore 0.85 :Window \@m{...} :BasisStrengths {...} :BasisPreferences {...}}
threat-detected	nil	{:Kind financial :Description "..." :Severity 0.7 :Horizon near :BasisWeaknesses {...}}

life-event	nil	{:EventType relocation :Description "... " :Disruption 0.6}
------------	-----	---

Situation percepts are typically synthesized: the Kojin's Analyst correlates Capability and Preference percepts against external data (often fetched through a sibling Aether Psyche on the same mind).

3.7 Channel Depth

The depth-psychological channel. Percepts here are *inferences*, always carrying :Confidence and :Evidence .

:Data	:Address	:Content (idiom)
ego-trait	nil	{:Trait "the competent provider" :Register ego :Confidence 0.8 :Evidence {...} :Stability stable}
shadow-trait	nil	{:Trait "resentment of obligation" :Register shadow :Confidence 0.55 :Evidence {avoidance-q89 projection-p3} :Stability emerging}
projection-detected	nil	{:Target "coworker Dan" :Attribution "lazy" :ShadowCandidate "own fear of idleness" :Confidence 0.5}
persona-gap	nil	{:Stated {...} :Revealed {...} :GapSize 0.6 :Domain work}
integration-signal	nil	{:Trait "resentment of obligation" :Direction acknowledged :Evidence {journal-entry-...}}

3.8 Channel System

:Data	:Address	:Content (idiom)
consent-change	nil	{:Scope medical :Granted no :Moment \@m{...}}

session-event	"kojin://alex/session-14"	{:EventType opened :SessionId "s14"}
connection-status	nil	{:Source gmail :Status connected :Error nil}

3.9 Percept Examples

```
[Percept :Modality Kojin :Channel Dialogue :Address
"kojin://alex/session-14"
  :Data utterance
  :Content {:Text "I turned down the promotion" :SessionId "s14"
    :Sentiment -0.3 :Topics {work}}
  :Moment \@m{...}]

[Percept :Modality Kojin :Channel Journal :Address nil
  :Data journal-pattern
  :Content {:PatternType recurring-theme :Theme "fear of being
ordinary"
    :Occurrences 7}
  :Moment \@m{...}]

[Percept :Modality Kojin :Channel Depth :Address nil
  :Data shadow-trait
  :Content {:Trait "resentment of obligation" :Register shadow
    :Confidence 0.55 :Evidence {avoidance-q89 projection-
p3}}
  :Moment \@m{...}]

[Percept :Modality Kojin :Channel Situation :Address "https://jobs..."
  :Data opportunity-detected
  :Content {:Kind career :FitScore 0.85 :BasisStrengths {public-
speaking}}
  :Moment \@m{...}]
```

4 Common :Content Idiom Slots

Slot	Type	Count	Notes
:Confidence	real	9	Certainty of the inference. Mandatory on all Depth and Capability percepts.
:Evidence	list	8	Provenance — the reifiers of the percepts that support this inference.
:Text / :Summary / :Description	string	9	The human-readable payload.
:Topics / :Domain / :Kind	list/literal	8	What area of life this concerns.
:Sentiment / :Valence / :Affect	real/literal	5	Emotional charge.
:Source	literal	4	stated , behavioral , inferred , or a channel set.
:SessionId / :ThreadId	string	4	Conversation context.
:Severity / :FitScore / :GapSize	real	3	Magnitude of the situation or gap.
:Sensitivity	literal	2	Governs storage and consent rules.
:Register	literal	2	ego or shadow — which side of the model.

The Kojin's signature slot pair is :Confidence + :Evidence : because so many Kojin percepts are inferences about an interior the mind can never observe directly, every inference must carry its certainty and its provenance.

5 Actuations

Actuations are how the mind reaches back to the person: questions, prompts, reflections, guidance. Following GIL convention, actuations are sent as ATTEMPT structures and results return as RESULT percepts — the mind learns what it said through perception.

5.1 Actuation Table

#	:Action	Description	:Parameters idiom
K01	say	Send a conversational message	{:Text "... " :SessionId "s14" :Tone warm}
K02	ask	Ask the person a question	{:Text "... " :QuestionId "q-90" :Topic work :Purpose elicit-preference}
K03	prompt-journal	Offer a journaling prompt	{:Text "... " :PromptId "p-13" :Theme obligation :Depth gentle}
K04	reflect	Mirror back a pattern	{:Pattern "... " :Evidence {...} :Framing tentative}
K05	summarize	Summarize a period or theme	{:Scope {\@m{...} \@m{...}} :Theme nil :Length brief}
K06	advise-opportunity	Surface an opportunity	{:OpportunityId "o-4" :Rationale "... " :Urgency 0.6}
K07	warn-threat	Surface a threat	{:ThreatId "t-2" :Rationale "... " :Severity 0.7 :Mitigations {...}}
K08	suggest-action	Recommend a concrete step	{:Action "... " :Basis {strengths preferences} :Effort low}
K09	ego-assert	Write a trait into the ego model	{:Trait "... " :Confidence 0.8 :Evidence {...}}
K10	shadow-assert	Write a trait into the shadow model	{:Trait "... " :Confidence 0.55 :Evidence {...}}
K11	model-retract	Remove a trait from either register	{:Trait "... " :Register shadow :Reason contradicted}
K12	preference-record	Record a preference	{:Domain food :Object "cilantro" :Valence -0.9 :Source stated}

K13	ingest-document	Request analysis of a shared file	{:Path "file:///uploads/x.pdf" :Depth full}
K14	consent-request	Ask permission for a data scope	{:Scope medical :Rationale "..."}
K15	schedule-checkin	Schedule a future conversation	{:At \@m{...} :Theme "the promotion decision"}

5.2 Actuation Domains

Conversation: say, ask, reflect, summarize

Journaling: prompt-journal

Guidance: advise-opportunity, warn-threat, suggest-action

Modeling: ego-assert, shadow-assert, model-retract, preference-record

Intake: ingest-document, consent-request

Continuity: schedule-checkin

5.3 ATTEMPT Examples

```

[ATTEMPT :Action ask
      :Parameters {:Text "When you imagine having taken the
promotion, what do you feel first?"
                  :QuestionId "q-90" :Topic work :Purpose depth-
probe}
      :Token ?tok]

[ATTEMPT :Action shadow-assert
      :Parameters {:Trait "resentment of obligation" :Confidence 0.55
                  :Evidence {avoidance-q89 projection-p3 journal-
pattern-7}}
      :Token ?tok]

[ATTEMPT :Action warn-threat
      :Parameters {:ThreatId "t-2" :Severity 0.7
                  :Rationale "spending trend exceeds income trend by
12%"
                  :Mitigations {...}}
      :Token ?tok]

```

6 Cross-Cutting Structure

6.1 The Four Tuple Types

```

PERCEPT:  [PERCEPT :Modality Kojin :Channel C :Address A :Data D
:Content {...} :Moment M :Token T]
ATTEMPT:    [ATTEMPT :Action A :Parameters {...} :Token T :By M]
RESULT:     [RESULT :Action A :Status S :Reason R :Moment M :Token T]
URGE:       [URGE :Need N :Source S :Delta D :Moment M :Token T]

```

6.2 Actuation → Percept Pairings

:Action	Produces :Data
---------	----------------

say , ask	utterance , question-answered , or question-avoided (eventual)
prompt-journal	journal-entry (eventual)
reflect	utterance — the person's reaction, sometimes integration-signal
advise-opportunity , warn-threat , suggest-action	utterance and later life-event
ego-assert , shadow-assert	ego-trait , shadow-trait (confirmation)
ingest-document	document-shared , record-shared
consent-request	consent-change
schedule-checkin	session-event (future)

Many Kojin percepts are spontaneous — unprompted messages, unsolicited journal entries, shared documents — arriving without a prior Attempt. Homeostatic signals arrive as URGE tuples: neglect (no contact beyond threshold), model-staleness (Depth model unrefreshed), unresolved-threat (warned threat with no mitigation observed), consent-gap (data present without covering consent).

7 The Jungian Model

The Kojin maintains a two-register model of the person in the Totality.

Ego register: the person as they know and present themselves — conscious identity, stated values, owned strengths and weaknesses, the narrative "I." Built primarily from stated material: Dialogue, Journal, Preference (:Source stated).

Shadow register: the person as they do not (yet) know themselves — disowned traits, avoided topics, projections onto others, gaps between stated and revealed preference, recurring dream figures, charged overreactions. Built from indirect evidence: question-avoided , projection-detected , persona-gap , journal-pattern , dream-report .

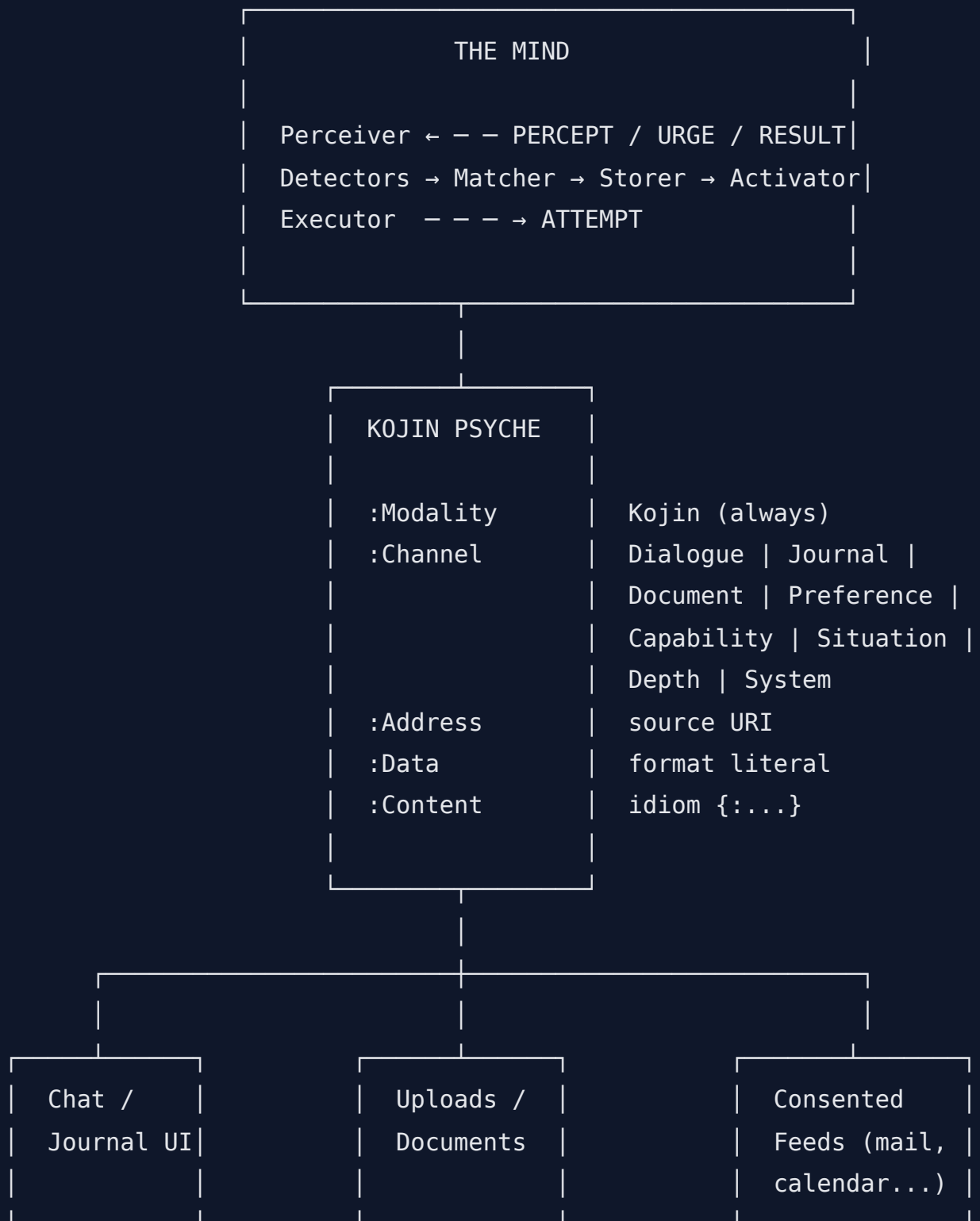
Three invariants govern the model:

1. **Evidence or nothing.** No trait enters either register without an `:Evidence` list pointing at stored percepts. Shadow traits additionally require evidence from at least two independent `:Data` types.
2. **Confidence decays.** Trait `:Confidence` decays over time unless refreshed by new evidence; contradicting evidence decrements it. Below the floor in `[Model]`, the trait is retracted.
3. **The shadow is offered, never imposed.** Shadow traits shape the mind's *questions* (`ask`, `prompt-journal`, `reflect` with `:Framing tentative`), not its *assertions to the person*. Only when an `integration-signal` shows the person acknowledging the trait themselves does it become citable in guidance.

The purpose is practical, not clinical: opportunities scored only against the ego model flatter; threats scored only against the ego model miss self-sabotage. Guidance grounded in both registers fits the whole person.

8 Architecture

8.1 The Kojin as Mediator



8.2 Two-Stage Perception

Raw material (an utterance, a journal entry, a document) produces a first-order percept immediately. The Analyst then runs over accumulated first-order percepts and emits second-order percepts — `journal-pattern`, `preference-revealed`, `strength-observed`, `persona-gap`, `ego-trait`, `shadow-trait`, `opportunity-detected`, `threat-detected`

— each citing its first-order evidence. The mind perceives both the person's life and the analysis of it through the same Percept relation.

9 Deployment File Set

File	Category	Description
<code>kojin.package</code>	Protocol	Session management, intake, analysis functions, model operations, consent registry
<code>kojin-psyche.theory</code>	Theory	Three agents: Listener, Analyst, Companion
<code>Kojin.daicho</code>	Configuration	Person identity, sources, consent scopes, analysis cadence, model floors, urge thresholds
<code>kojin.suites</code>	Tests	9 suites: intake, dialogue, journaling, preferences, capabilities, depth model, guidance, consent, full lifecycle

10 The Package

`kojin.package` provides:

Sessions: `kojin-session-open`, `kojin-session-close`, `kojin-say`, `kojin-ask`.

Intake: `kojin-ingest-text`, `kojin-ingest-document`, `kojin-ingest-feed`.

Analysis: `kojin-analyze-sentiment`, `kojin-analyze-topics`, `kojin-find-patterns`, `kojin-detect-avoidance`, `kojin-detect-projection`, `kojin-score-fit` (opportunities), `kojin-score-risk` (threats).

Model: `kojin-ego-assert`, `kojin-shadow-assert`, `kojin-model-retract`, `kojin-model-decay`, `kojin-preference-record`, `kojin-model-snapshot`.

Consent: `kojin-consent-grant`, `kojin-consent-revoke`, `kojin-consent-check` — every intake and analysis function calls `kojin-consent-check` before touching data in a scoped category.

Dispatch: `kojin-dispatch` maps any `Kojin :Action` literal to the corresponding function.

11 The Theory

`kojin-psyche.theory` defines three agents:

- Listener** — reactive. Receives raw material from the person (messages, journal entries, uploads), wraps each as a first-order PERCEPT, and tells the Perceiver.
- Analyst** — proactive, slow interval (default 1440 Moments — daily). Runs the analysis functions over accumulated first-order percepts, emits second-order PERCEPTs (patterns, preferences, capabilities, Depth traits, opportunities, threats), applies model decay, and monitors homeostatic conditions, emitting URGE tuples for `neglect` , `model-staleness` , `unresolved-threat` , and `consent-gap` .
- Companion** — reactive. Receives ATTEMPTs from the Executor, dispatches through `kojin-dispatch` , wraps the outcome as a RESULT tuple, and tells the Perceiver.
-

12 Configuration

`Kojin.daicho` contains six sections:

Section	Contents
[Psyche]	Agent URL, port, delay, modality literal
[Person]	Person identifier, display name, locale, timezone
[Sources]	Chat/journal UI endpoint, upload directory, consented feed URIs
[Consent]	Scoped categories (general, financial, medical, relational, dreams) and grant status
[Model]	Confidence floor, decay rate, shadow evidence minimum, analysis cadence
[Urges]	Thresholds for neglect, model-staleness, unresolved-threat, consent-gap

13 Test Suites

`kojin.suites` contains 9 test suites:

Suite	Tests
<code>Kojin.intake</code>	Text, document, and feed ingestion; first-order percept formation
<code>Kojin.dialogue</code>	Session lifecycle, say/ask round-trips, avoidance detection
<code>Kojin.journaling</code>	Prompting, entry ingestion, pattern detection across entries
<code>Kojin.preferences</code>	Stated vs revealed recording, conflict surfacing
<code>Kojin.capabilities</code>	Strength/weakness inference with evidence chains
<code>Kojin.depth-model</code>	Ego/shadow assert, evidence requirements, decay, retraction, integration
<code>Kojin.guidance</code>	Opportunity fit scoring, threat risk scoring, advise/warn dispatch
<code>Kojin.consent</code>	Grant, revoke, scope enforcement on every intake path
<code>Kojin.full-lifecycle</code>	End-to-end: open, converse, journal, ingest, analyze, model, guide, close

14 Privacy and Consent

The Kojin's world is a person; the ethics are therefore part of the specification, not an afterthought.

Consent is scoped and revocable. Data categories in `[Consent]` gate every intake and analysis path. Revocation stops future intake in that scope and marks existing scoped percepts inert.

The person can see the model. `kojin-model-snapshot` renders both registers, with evidence, on request. Nothing about the person is withheld from the person — though shadow material is *presented* per invariant 3 of § 7 (offered through questions, not pronounced).

The token is the person's. The `:Token` on every PERCEPT, RESULT, and URGE tuple is the person's credential, registered with the GIL Registrar at startup. One Kojin, one human.

15 Domain Coverage

Domain	:Channel	Percept :Data values	Actuations
Conversation	Dialogue	utterance , question-answered , question-avoided	say , ask , reflect , summarize
Reflection	Journal	journal-entry , journal-pattern , dream-report	prompt-journal
Shared Material	Document	document-shared , message-shared , record-shared	ingest-document
Wants	Preference	preference-signal , preference-revealed , value-signal	preference-record
Abilities	Capability	strength-observed , weakness-observed , growth-signal	— (inferred)
Circumstances	Situation	opportunity-detected , threat-detected , life-event	advise-opportunity , warn-threat , suggest-action
Interior	Depth	ego-trait , shadow-trait , projection-detected , persona-gap , integration-signal	ego-assert , shadow-assert , model-retract
Governance	System	consent-change , session-event , connection-status	consent-request , schedule-checkin

16 Etymology

Kojin (Japanese 個人, *kojin*) — a compound of 個 (*ko*), "individual, the discrete unit, the counter for single distinct things," and 人 (*jin*), "person, human being." The word means the individual person — the private self as distinct from the group, the public, the role. Japanese

law and philosophy use *kojin* precisely where English uses "the individual": 個人情報 (*kojin jōhō*) is "personal data," and 個人主義 (*kojin shugi*) is "individualism."

The compound is exact for this Psyche twice over. First, 個 insists on *one*: the Kojin interfaces with a single human being, one world per mind, one token per person. Second, *kojin jōhō* — personal data — is literally what flows through it.

Jung's own terms echo in the design: the *ego* is the "I" the person knows; the *shadow* (*Schatten*) is, in Jung's phrase, "the thing a person has no wish to be." The Kojin holds both registers so that guidance addresses the whole individual — the 個人 — and not merely the persona.

Where Aether names the medium (propagation), Expanse names the territory (vastness), and Eidos names the act of seeing (knowing), Kojin names the subject itself — the one person who is this Psyche's entire world.

Shudan (集団, "the group") is the sibling Psyche: the same architecture turned toward an organization-as-world. Kojin is the individual; Shudan is the collective. Together they cover the two forms of human world a GIL mind can accompany.